

Smart City Emergency Response using BLYNK

Project Aim

To develop a Smart City Emergency Response and Safety Management System using ESP32 and Blynk IoT.

Project Overview

The Smart City Emergency Response and Safety Management System monitors parking occupancy, air quality, temperature, and emergency situations using various sensors. It automatically activates alarms, emergency barriers, and safety devices while displaying real-time city status on an OLED screen and Blynk IoT platform.

Components Used

- ESP32 Development Board
- HC-SR04 Ultrasonic Sensor
- MQ2 Gas Sensor
- DHT22 Temperature Sensor
- SSD1306 OLED Display
- Push Button
- Green LED
- Red LED
- Buzzer
- Servo Motor
- Relay Module
- 220 Ω Resistors (2)
- Jumper Wires
- Blynk

Construction and Wiring Details

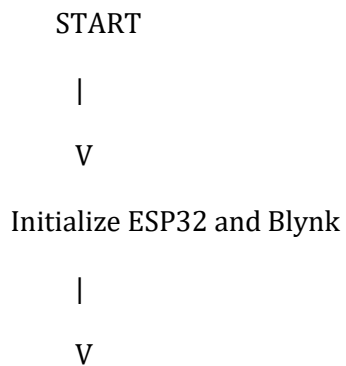
Component	Pin	ESP32 Pin
HC-SR04	TRIG	GPIO 5
HC-SR04	ECHO	GPIO 18
MQ2 Gas Sensor AO		GPIO 34
MQ2 Gas Sensor VCC		5V
MQ2 Gas Sensor GND		GND
DHT22	DATA	GPIO 4
DHT22	VCC	3.3V
DHT22	GND	GND
OLED Display	SDA	GPIO 21
OLED Display	SCL	GPIO 22
OLED Display	VCC	3.3V
OLED Display	GND	GND
Push Button	One Side	GPIO 27
Push Button	Other Side	GND
Green LED	Anode	GPIO 12
Red LED	Anode	GPIO 14
LEDs Cathode	Through 220Ω Resistor GND	
Buzzer (+)	GPIO 26	
Buzzer (-)	GND	
Servo Signal	GPIO 13	

Component	Pin	ESP32 Pin
Servo VCC	5V	
Servo GND	GND	
Relay IN	GPIO 25	
Relay VCC	5V	
Relay GND	GND	

Blynk Connection

DataStream Name	Virtual Pin	Data Type	Min	Max
Parking Status	V0	String	-	-
Gas Value	V1	Integer	0	1000
Temperature	V2	Double	0	80
Barrier Status	V3	String	-	-
City Status	V4	String	-	-
Relay Status	V6	Integer	0	1

Code Structure / Program Flow



Read Ultrasonic Sensor

|

Distance < 10 cm ?

/ \

Yes

No

| |

Parking Full

Parking Available

Red LED ON

Green LED ON

|

V

Read MQ2 Sensor

|

Gas Value > 600 ?

/ \

Yes

No

| |

Pollution Alert

Air Safe

Buzzer ON

|

V

Read DHT22 Sensor

|

Temperature > 35°C ?

/ \

Yes

No

| |

Relay ON Normal State

Buzzer ON

|

V

Emergency Button Pressed?

/ \

Yes No

| |

Servo 90° Servo 0°

Buzzer ON

|

V

Gas Alert AND Fire Alert ?

/ \

Yes No

| |

EMERGENCY WARNING

Servo 90°

Relay ON

Red LED ON

Buzzer ON

|

V

Update OLED Dashboard

|

V

Send Data to Blynk

|

V

REPEAT

Source Code

Diagram.jetson

```
{  
  "version": 1,  
  "author": "Aman Abdulla",  
  "editor": "wokwi",  
  "parts": [  
    { "type": "board-esp32-devkit-c-v4", "id": "esp", "top": 48, "left": 24.04, "attrs": {} },  
    {  
      "type": "wokwi-hc-sr04",  
      "id": "ultrasonic1",  
      "top": -36.9,  
      "left": 159.1,  
      "attrs": { "distance": "81" }  
    },  
    { "type": "wokwi-gas-sensor", "id": "gas1", "top": -26.1, "left": -175.4, "attrs": {} },  
    {  
      "type": "wokwi-dht22",  
      "id": "dht1",  
      "top": 57.9,  
      "left": 282.6,
```

```
"attrs": { "temperature": "-10.9" }  
},  
{  
  "type": "board-ssd1306",  
  "id": "oled1",  
  "top": 300.74,  
  "left": 29.03,  
  "attrs": { "i2cAddress": "0x3c" }  
},  
{  
  "type": "wokwi-pushbutton",  
  "id": "btn1",  
  "top": 335.6,  
  "left": -89.2,  
  "rotate": 270,  
  "attrs": { "color": "green", "xray": "1" }  
},  
{  
  "type": "wokwi-led",  
  "id": "led1",  
  "top": 82.8,  
  "left": -245.8,  
  "attrs": { "color": "green" }  
},  
{  
  "type": "wokwi-led",
```

```
"id": "led2",  
  
"top": 82.8,  
  
"left": -188.2,  
  
"attrs": { "color": "red" }  
},  
  
{  
  
  "type": "wokwi-resistor",  
  
  "id": "r1",  
  
  "top": 158.4,  
  
  "left": -192.55,  
  
  "rotate": 90,  
  
  "attrs": { "value": "220" }  
},  
  
{  
  
  "type": "wokwi-resistor",  
  
  "id": "r2",  
  
  "top": 158.4,  
  
  "left": -250.15,  
  
  "rotate": 90,  
  
  "attrs": { "value": "220" }  
},  
  
{  
  
  "type": "wokwi-buzzer",  
  
  "id": "bz1",  
  
  "top": 223.2,  
  
  "left": 289.8,
```



```
"attrs": { "volume": "0.1" }
},
{
  "type": "wokwi-servo",
  "id": "servo1",
  "top": 313,
  "left": -277.4,
  "rotate": 90,
  "attrs": {}
},
{ "type": "wokwi-relay-module", "id": "relay1", "top": 57.8, "left": -124.8, "attrs": {} }
],
"connections": [
  [ "esp:TX", "$serialMonitor:RX", "", [] ],
  [ "esp:RX", "$serialMonitor:TX", "", [] ],
  [ "esp:18", "ultrasonic1:ECHO", "green", [ "h0" ] ],
  [ "ultrasonic1:TRIG", "esp:5", "green", [ "v0" ] ],
  [ "esp:GND.2", "ultrasonic1:GND", "black", [ "v0", "h134.4" ] ],
  [ "esp:5V", "ultrasonic1:VCC", "red", [ "h-9.45", "v19.2", "h211.2" ] ],
  [ "esp:34", "gas1:AOUT", "green", [ "h-28.65", "v-124.8" ] ],
  [ "gas1:VCC", "esp:5V", "red", [ "h19.2", "v229.5" ] ],
  [ "gas1:GND", "esp:GND.2", "black", [ "h182.4", "v66.4" ] ],
  [ "dht1:SDA", "esp:4", "green", [ "v0" ] ],
  [ "dht1:VCC", "esp:3V3", "red", [ "v0", "h-153.6", "v-172.8", "h-134.4", "v76.8" ] ],
  [ "esp:GND.2", "dht1:GND", "black", [ "v0", "h86.4", "v134.4", "h115.2" ] ],
  [ "esp:21", "oled1:SDA", "green", [ "h38.4", "v163.2", "h-67.2" ] ],
```

```

[ "esp:22", "oled1:SCL", "green", [ "h48", "v192", "h-86.4" ] ],
[ "oled1:GND", "esp:GND.1", "black", [ "v-19.2", "h-76.8", "v-9.6" ] ],
[ "esp:3V3", "oled1:VCC", "green", [ "h-67.05", "v201.6", "h115.2" ] ],
[ "led1:A", "r2:1", "green", [ "v0" ] ],
[ "led2:A", "r1:1", "green", [ "v0" ] ],
[ "esp:GND.1", "led2:C", "black", [ "h-201.45" ] ],
[ "esp:GND.1", "led1:C", "black", [ "h-268.65" ] ],
[ "esp:12", "r2:2", "green", [ "v9.6", "h-249.45" ] ],
[ "esp:14", "r1:2", "green", [ "h-172.65", "v9.6", "h9.6" ] ],
[ "esp:13", "servo1:PWM", "green", [ "h-220.65" ] ],
[ "servo1:GND", "esp:GND.1", "black", [ "v0" ] ],
[ "esp:5V", "servo1:V+", "red", [ "h-134.25", "v-28.8", "h-76.8" ] ],
[ "relay1:IN", "esp:25", "green", [ "h0", "v47.8" ] ],
[ "relay1:GND", "esp:GND.1", "black", [ "h-9.6", "v114.8" ] ],
[ "esp:5V", "relay1:VCC", "red", [ "h-172.65", "v-172.8" ] ],
[ "esp:27", "btn1:2.r", "green", [ "h0" ] ],
[ "esp:GND.1", "btn1:1.r", "black", [ "h-105.45" ] ],
[ "esp:26", "bz1:2", "green", [ "h-9.45", "v124.8", "h230.4", "v28.8" ] ],
[ "esp:GND.3", "bz1:1", "black", [ "h76.8", "v211.2", "h105.6" ] ]
],
"dependencies": {}
}

```

Esp32 Code

```
#define BLYNK_TEMPLATE_ID "TMPL3iDk4fk_A"

#define BLYNK_TEMPLATE_NAME "Smart City"

#define BLYNK_AUTH_TOKEN "4Q2Vrwiqmw9ZUI5-6qvsu-bSBgjxOYwH"


#include <WiFi.h>

#include <BlynkSimpleEsp32.h>

#include <DHT.h>

#include <Wire.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>

#include <ESP32Servo.h>


char ssid[] = "Wokwi-GUEST";

char pass[] = "";


#define TRIG_PIN 5

#define ECHO_PIN 18

#define MQ2_PIN 34

#define DHT_PIN 4

#define BUTTON_PIN 27


#define GREEN_LED 12

#define RED_LED 14

#define BUZZER_PIN 26
```

```
#define RELAY_PIN 25
```

```
#define SERVO_PIN 13
```

```
#define DHTTYPE DHT22
```

```
DHT dht(DHT_PIN, DHTTYPE);
```

```
Servo barrierServo;
```

```
#define SCREEN_WIDTH 128
```

```
#define SCREEN_HEIGHT 64
```

```
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
```

```
String parkingStatus;
```

```
String gasStatus;
```

```
String barrierStatus;
```

```
String cityStatus;
```

```
void setup() {
```

```
  Serial.begin(115200);
```

```
  pinMode(TRIG_PIN, OUTPUT);
```

```
  pinMode(ECHO_PIN, INPUT);
```

```
  pinMode(GREEN_LED, OUTPUT);
```

```
  pinMode(RED_LED, OUTPUT);
```

```
pinMode(BUTTON_PIN, INPUT_PULLUP);
```

```
pinMode(BUZZER_PIN, OUTPUT);
```

```
pinMode(RELAY_PIN, OUTPUT);
```

```
barrierServo.attach(SERVO_PIN);
```

```
barrierServo.write(0);
```

```
dht.begin();
```

```
Wire.begin(21,22);
```

```
display.begin(SSD1306_SWITCHCAPVCC,0x3C);
```

```
display.clearDisplay();
```

```
display.setTextSize(1);
```

```
display.setTextColor(WHITE);
```

```
Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
```

```
}
```

```
void loop() {
```

```
Blynk.run();
```

```
//===== Parking System =====
```

```
digitalWrite(TRIG_PIN, LOW);
```

```
delayMicroseconds(2);
```

```
digitalWrite(TRIG_PIN, HIGH);
```

```
delayMicroseconds(10);
```

```
digitalWrite(TRIG_PIN, LOW);
```

```
long duration = pulseIn(ECHO_PIN, HIGH);
```

```
float distance = duration * 0.034 / 2;
```

```
if(distance < 10)
```

```
{
```

```
    parkingStatus = "FULL";
```

```
    digitalWrite(RED_LED, HIGH);
```

```
    digitalWrite(GREEN_LED, LOW);
```

```
}
```

```
else
```

```
{
```

```
    parkingStatus = "AVAILABLE";
```

```
    digitalWrite(GREEN_LED, HIGH);
```

```
    digitalWrite(RED_LED, LOW);
```

```
}
```

```
//===== MQ2 =====
```

```
int gasRaw = analogRead(MQ2_PIN);
```

```
int gasValue = map(gasRaw,0,4095,0,1000);
```

```
bool gasAlert = false;
```

```
if(gasValue > 600)
```

```
{
```

```
    gasStatus = "POLLUTION ALERT";
```

```
    gasAlert = true;
```

```
}
```

```
else
```

```
{
```

```
    gasStatus = "SAFE";
```

```
}
```

```
//===== Temperature =====
```

```
float temperature = dht.readTemperature();
```

```
bool fireAlert = false;
```

```
if(temperature > 35)
```

```
{
```

```
    fireAlert = true;
}

//===== Emergency Button =====

bool buttonAlert = false;

if(digitalRead(BUTTON_PIN)==LOW)
{
    buttonAlert = true;
}

//===== Combined Emergency =====

bool cityEmergency = gasAlert && fireAlert;

if(buttonAlert || cityEmergency)
{
    cityStatus = "EMERGENCY";
}
else if(gasAlert || fireAlert)
{
    cityStatus = "WARNING";
}
else
{

```



```
cityStatus = "NORMAL";  
}
```

```
//===== Relay =====
```

```
if(fireAlert || cityEmergency)  
{  
    digitalWrite(RELAY_PIN,HIGH);  
}  
else  
{  
    digitalWrite(RELAY_PIN,LOW);  
}
```

```
//===== Servo Barrier =====
```

```
if(buttonAlert || cityEmergency)  
{  
    barrierServo.write(90);  
    barrierStatus = "OPEN";  
}  
else  
{  
    barrierServo.write(0);  
    barrierStatus = "CLOSED";  
}
```

```
//===== Buzzer =====
```

```
if(gasAlert || fireAlert || buttonAlert || cityEmergency)
```

```
{
```

```
    digitalWrite(BUZZER_PIN,HIGH);
```

```
}
```

```
else
```

```
{
```

```
    digitalWrite(BUZZER_PIN,LOW);
```

```
}
```

```
//===== OLED =====
```

```
display.clearDisplay();
```

```
display.setCursor(0,0);
```

```
display.println("SMART CITY");
```

```
display.print("Parking:");
```

```
display.println(parkingStatus);
```

```
display.print("Gas:");
```

```
display.println(gasStatus);
```

```
display.print("Temp:");
```

```
display.print(temperature);
```

```
display.println(" C");
```

```
display.print("Barrier:");
```

```
display.println(barrierStatus);
```

```
display.print("Status:");
```

```
display.println(cityStatus);
```

```
display.display();
```

```
//===== Blynk =====
```

```
Blynk.virtualWrite(V0, parkingStatus);
```

```
Blynk.virtualWrite(V1, gasValue);
```

```
Blynk.virtualWrite(V2, temperature);
```

```
Blynk.virtualWrite(V3, barrierStatus);
```

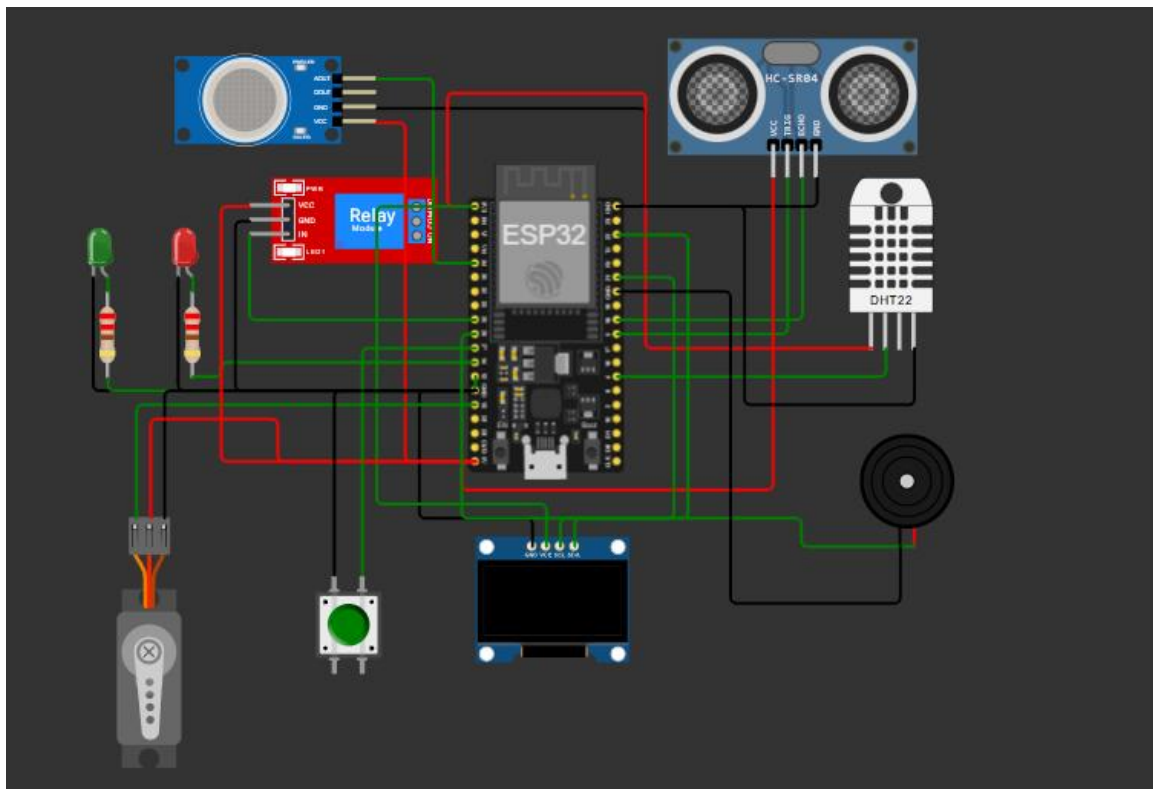
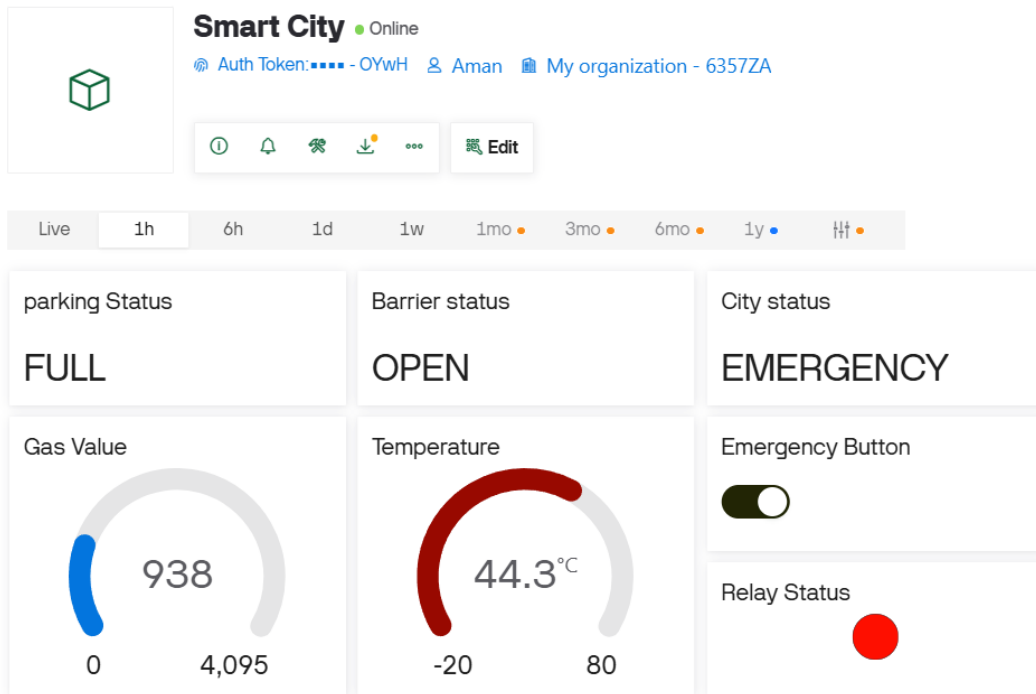
```
Blynk.virtualWrite(V4, cityStatus);
```

```
Blynk.virtualWrite(V6, digitalRead(RELAY_PIN));
```

```
delay(500);
```

```
}
```

Project Photograph



Links

Google drive :

<https://drive.google.com/drive/folders/1Fh3fHSctNxztdWg1exrqggivCa9gcN9A?usp=sharing>

Wokwi: <https://wokwi.com/projects/466683259092713473>

Git hub repo : <https://github.com/AmanAZ5950/CIRCUITRON.git>